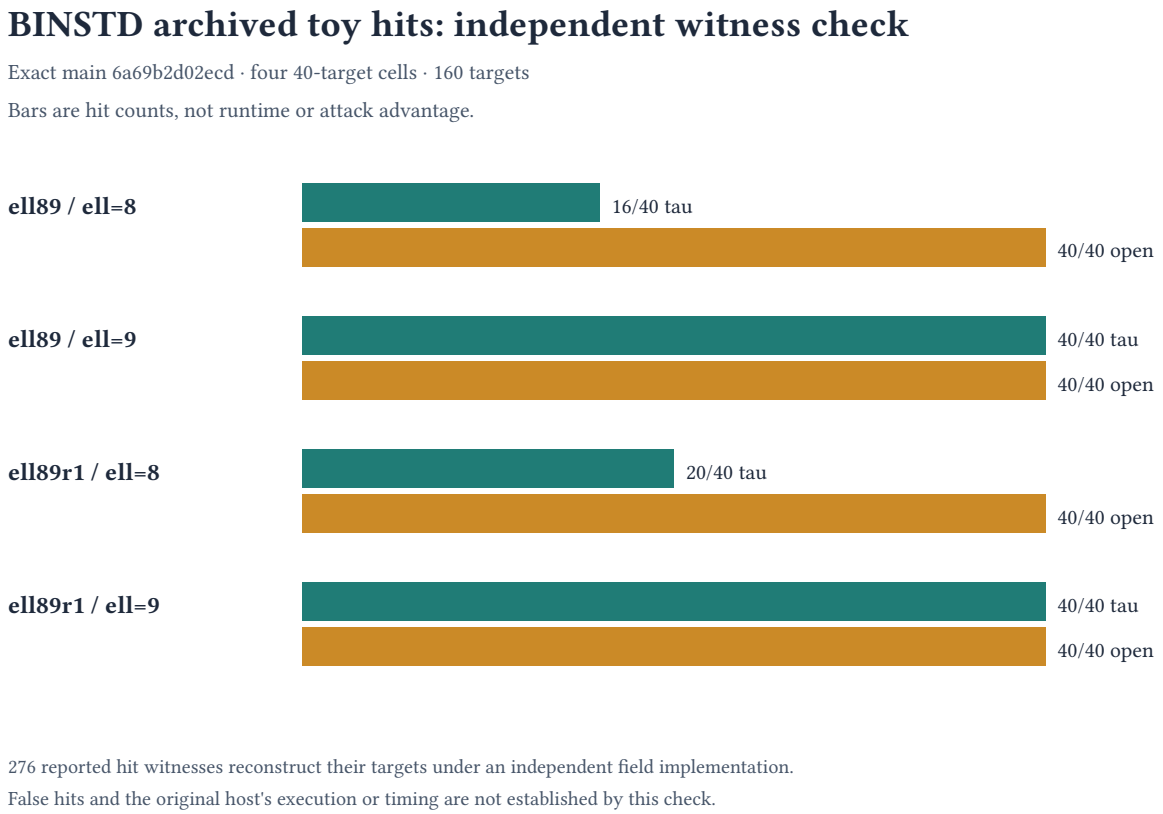
BINSTD archived toy witness audit

26 October 2026 · crypto-autoresearcher main `6a69b2d02ecd53f1649bcf56e23e4ffe91d953ad`

This is a source-bound audit of four archived `n=17`, `ell=8/9` cells. It adds no run or timing measurement.

Inputs and independent check

The four frozen sheriff per-target receipts name 160 targets and 276 positive hit witnesses. `check.py` rehashes six source/archive paths per receipt at their historical commits; it checks receipt, raw-result and yield-matrix bindings. It verifies all targets and witness triples on the curve $y^2+xy=x^3+1$ using independent interleaved field multiplication and Euclid inversion, and rejects an on-curve witness mutation in each cell. The addition formula is checked against SEC 1 v2 §2.2.2.



Observation

Cell	Tau	Open	Witnesses
ell89 / ell 8	16/40	40/40	56
ell89 / ell 9	40/40	40/40	80
ell89r1 / ell 8	20/40	40/40	60
ell89r1 / ell 9	40/40	40/40	80

All archived count pairs satisfy the premises of the **conditional** Lean `YieldCeiling.saturated_open_below_band` theorem. Exhaustive finite trial division independently checks that the archived `t^17+t^3+1` modulus and degree-eight factors are irreducible, and their product is `X^17+1`. This is computed evidence, not an added Lean proof.

Boundary and next check

The checker validates positive witnesses but does not independently prove every reported miss. No original per-target execution trace or host timing certificate is established. The normal-basis/Frobenius bridge is also not formalized. Obtain the original trace or execution proof; then add a pinned Lean irreducibility and model-bridge proof with axiom and semantic review.

Source: `research/binstd_end_to_end_20261026/REPORT.md`, `evidence.json`, and SEC 1 v2 §2.2.2 (https://www.secg.org/sec1-v2.pdf).